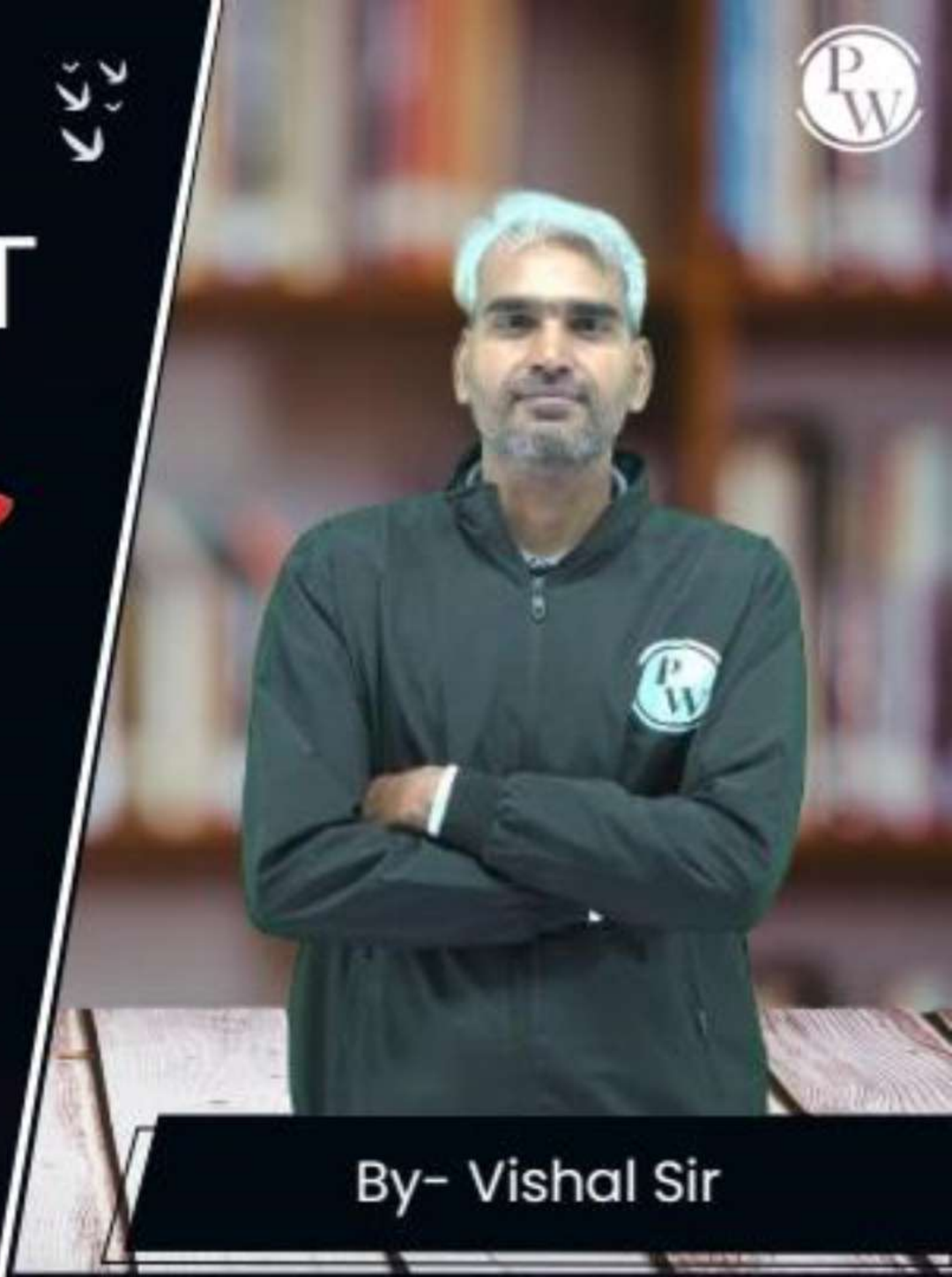
Computer Science & IT



Database Management System

Transaction & concurrency control

Lecture No. 04



By- Vishal Sir

Topics to be Covered



- Topic: Problems because of concurrent execution
- Topic: RW problem
- Topic: WR problem
- Topic: WW problem
- Topic: Lost update problem
- Topic: Classification of schedule

Slide

Recap of Previous Lecture



- Topic: Consistency
  - Schedule must be strict recoverable
  - Schedule must be serializable
- Topic: Equivalent schedule
- Topic: Serializable schedule

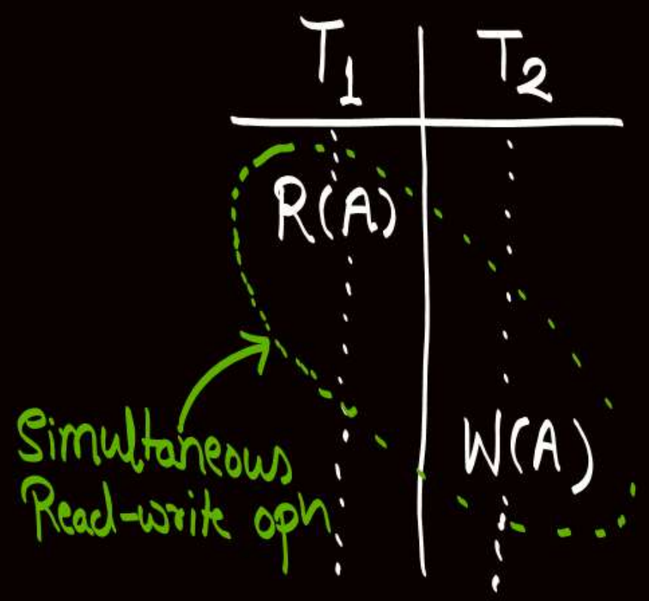
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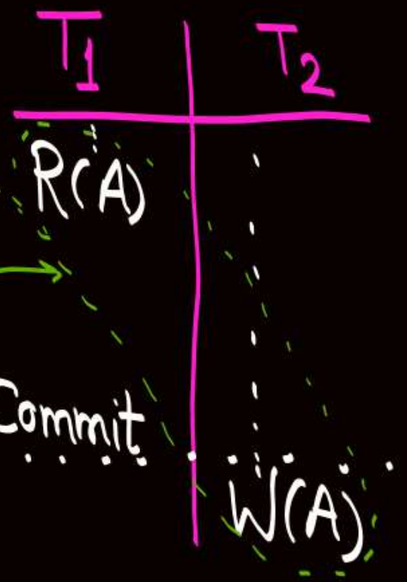
Topic : Problems because of concurrent execution

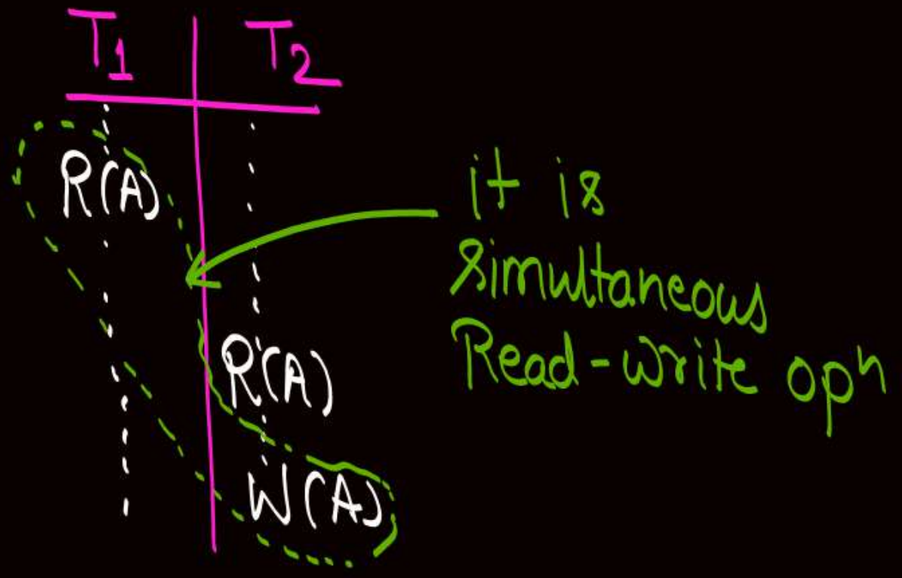
1. R W Problem / Read-Write Problem / Write after Read problem
2. W R problem / Write–Read Problem / Read after Write problem / Dirty Read Problem
3. W W Problem / Write-Write Problem / Write-after-Write problem
4. Lost update problem

RW Problem / Read-Write Problem / Write after Read Problem



* If transaction T_2 updates the dataitem 'A' which is already read by an uncommitted transaction T_1, then that sequence of operation is called Simultaneous Read-write operation







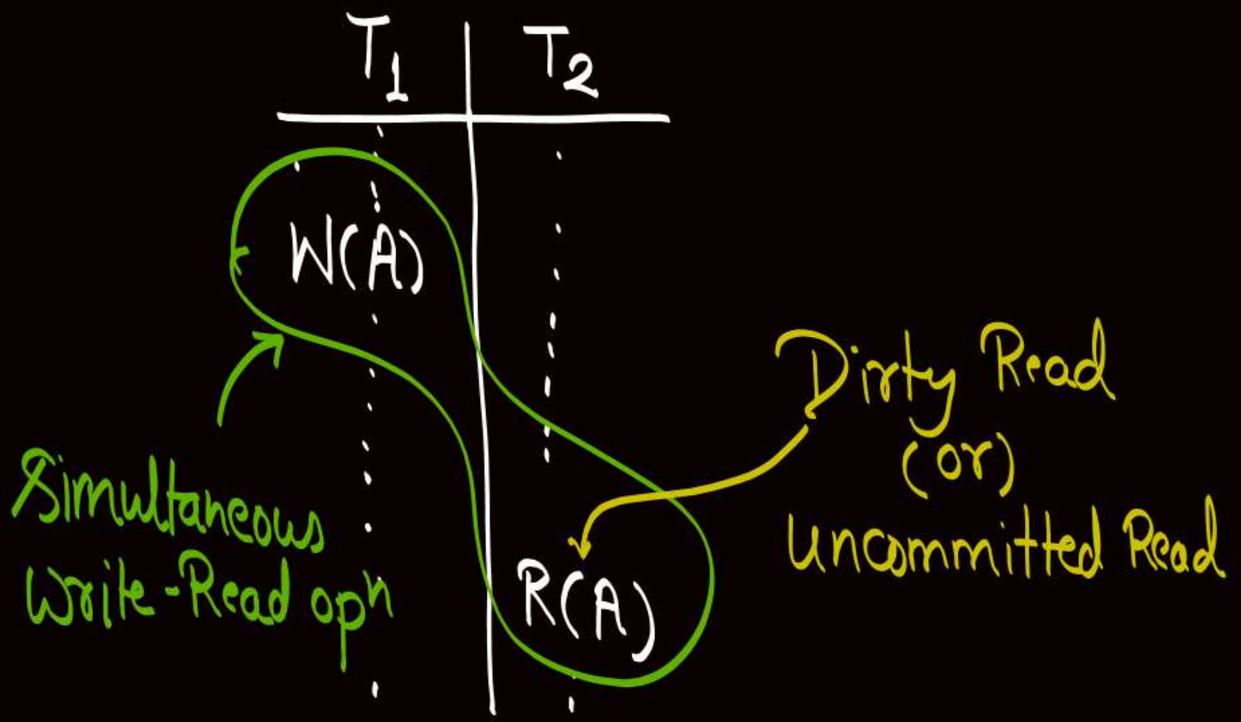
Topic : RW problem

A schedule is said to have RW problem if and only if,

① Simultaneous Read-write opn should exist

and ② Schedule must be a non-serializable schedule

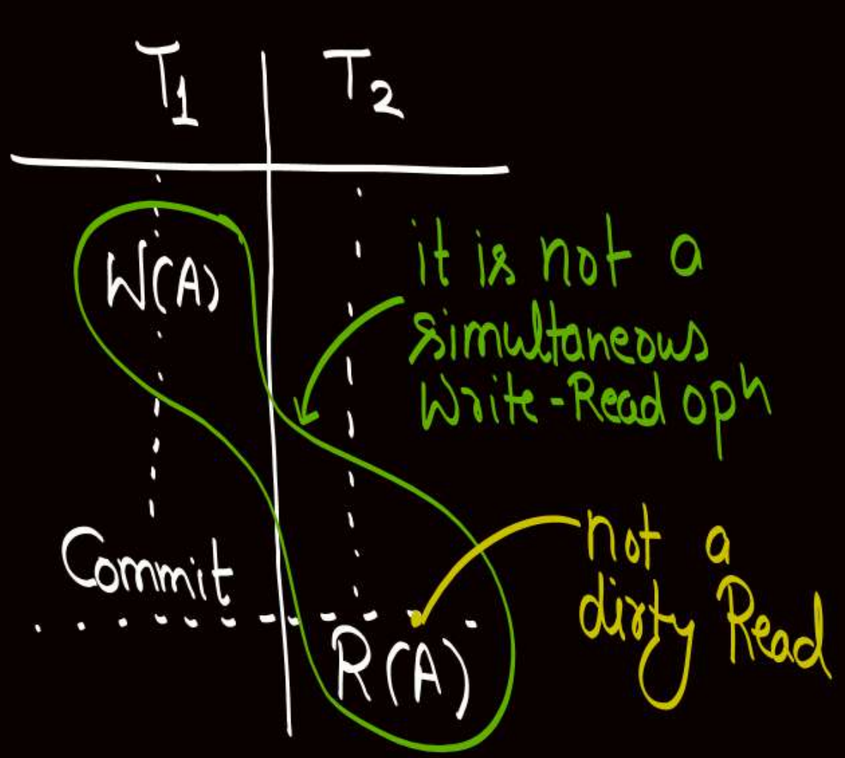
WR Problem / Write-Read Problem / Read after Write Problem
(or)
Dirty Read Problem

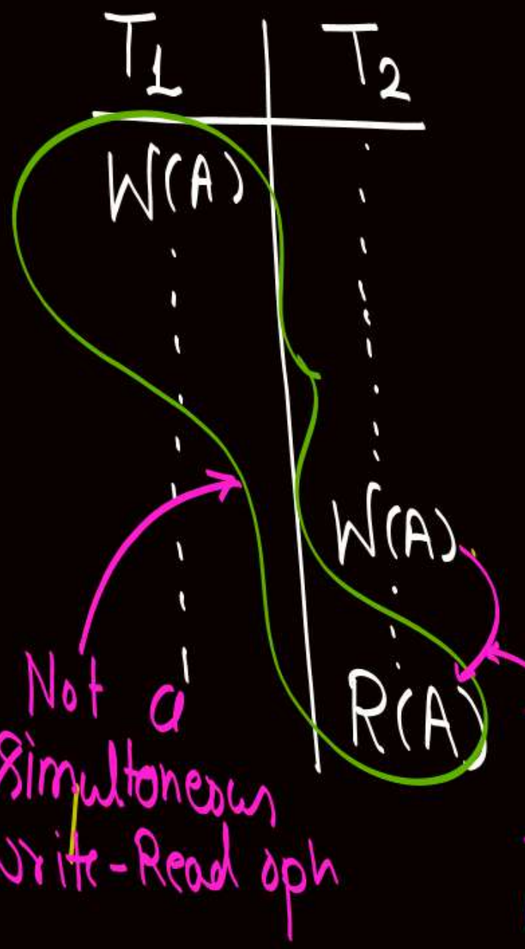


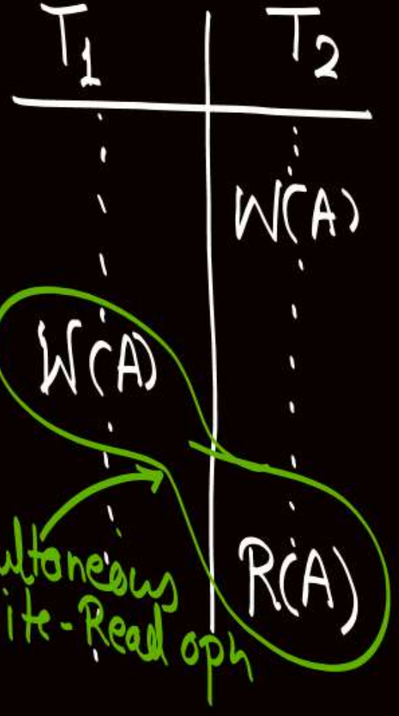
If transaction T_2 reads the value of dataitem 'A' updated by uncommitted transaction T_1, then that sequence of operation is called Simultaneous write-Read oph.

And that read oph is called "dirty Read" or "uncommitted Read"

WR Problem / Write-Read Problem / Read after Write Problem (or) Dirty Read Problem









Topic : WR problem

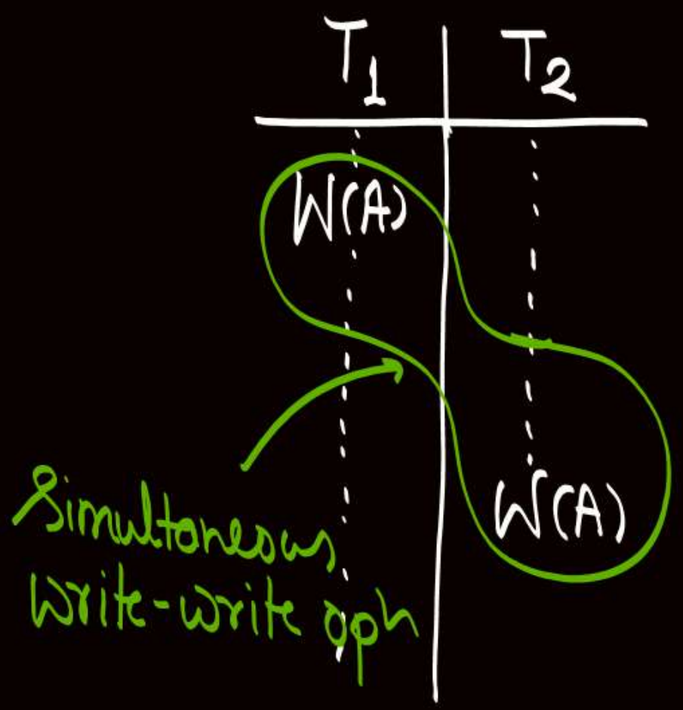
/ Read after Write Problem / Dirty Read problem

* A schedule is said to have WR problem if and only if,

① Simultaneous Write-Read op^{n} should exist
{i.e. Uncommitted Read op^{n} must exist}

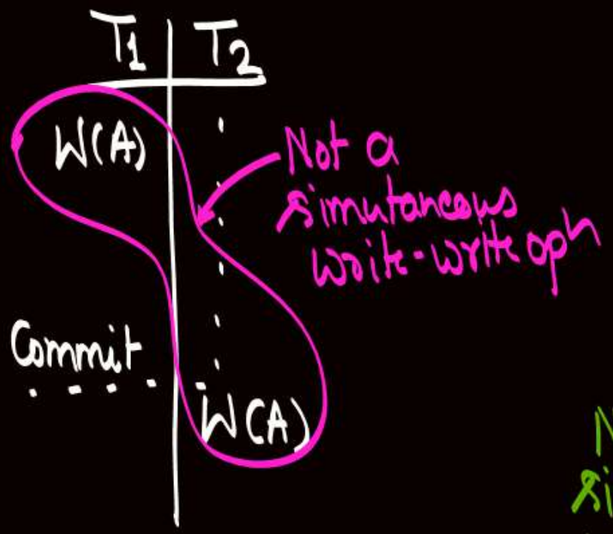
and ② Schedule must be a non-serializable schedule.

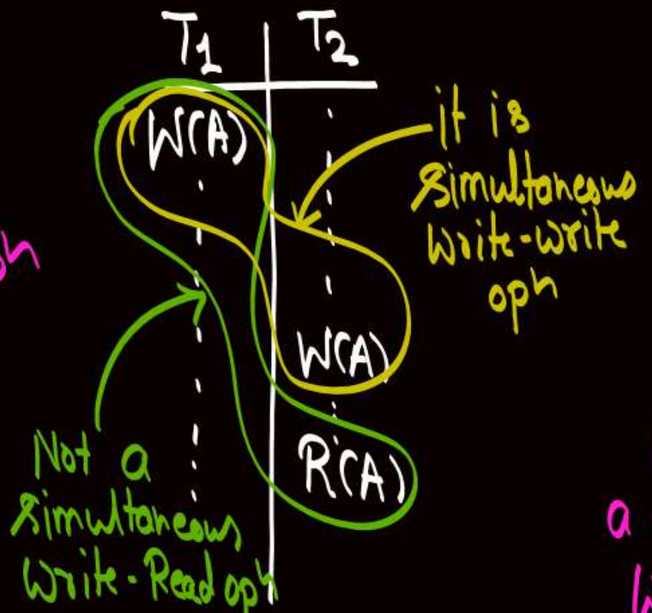
WW Problem / Write-Write Problem / Write after Write Problem

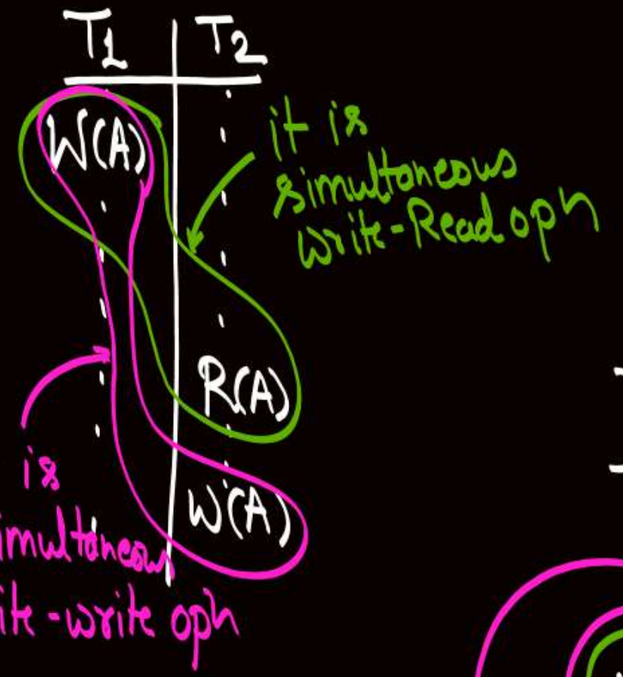


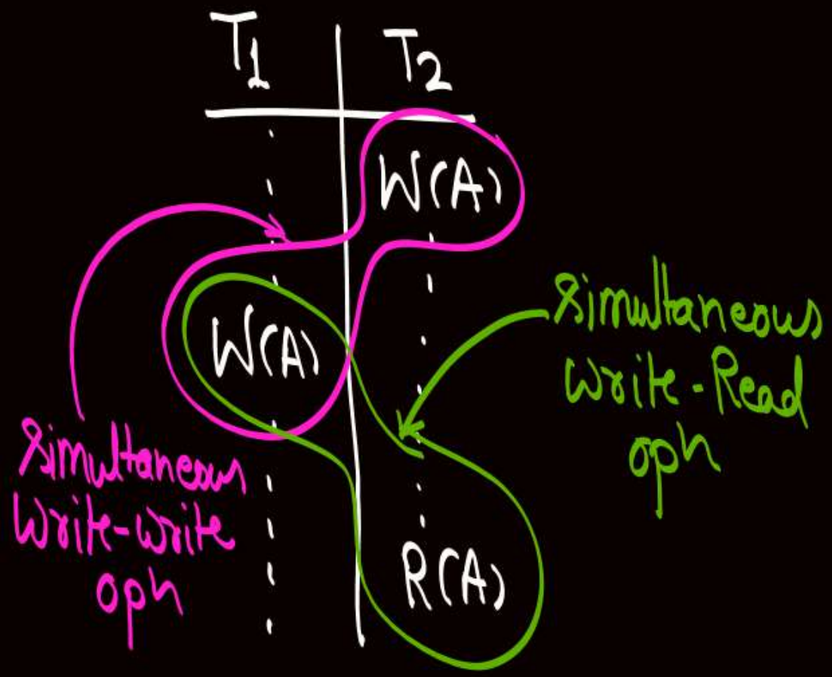
* If transaction T_2 updates the dataitem A which is already updated by an uncommitted transaction T_1, then that sequence of operation is called simultaneous write-write oph.

WW Problem / Write-Write Problem / Write after Write Problem











Topic : WW problem

- A schedule is said to have WW problem if and only if,

① Simultaneous Write-Write oph should exist

and ② Schedule must be a non-serializable schedule



Topic : Problems because of concurrent execution

1. RW Problem / Read-Write Problem / Write after Read problem
2. WR problem / Write-Read Problem / Read after write problem / Dirty Read Problem
3. WW Problem / Write-Write Problem / Write-after-write problem
4. Lost update problem

← Lost update problem is possible with Serializable as well as non-serializable schedules

These problems are possible only if schedule is non-serializable

Note:- If schedule is serializable schedule, then none of RW Problem, WR problem or WW problem can exist in the schedule, but lost update problem is still possible

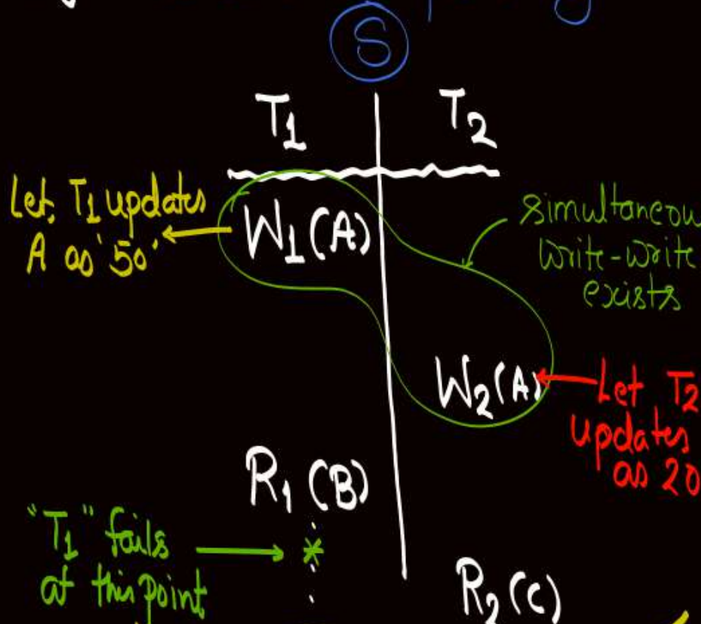


Topic : Lost update problem

A schedule is said to have lost-update problem if and only if simultaneous Write-Write op^n exists in the schedule. {does not matter whether the schedule is serializable or non-serializable}

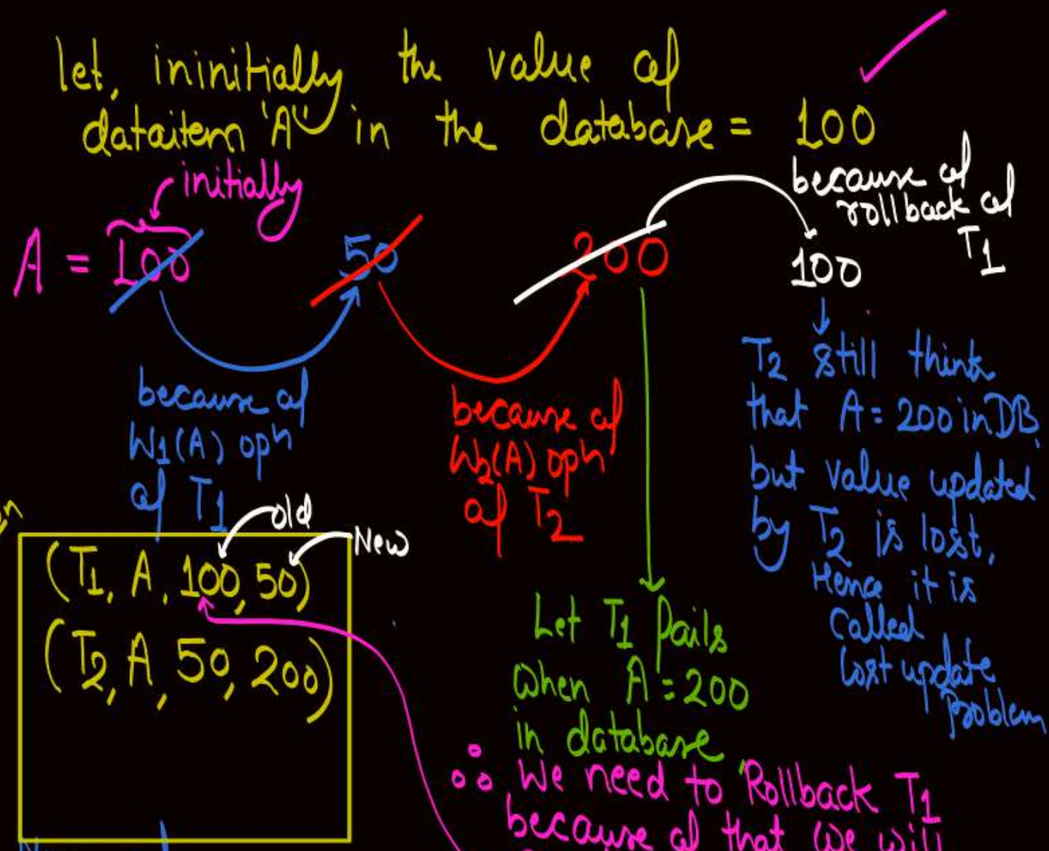
Lost-update problem

eg: Consider the following schedule S



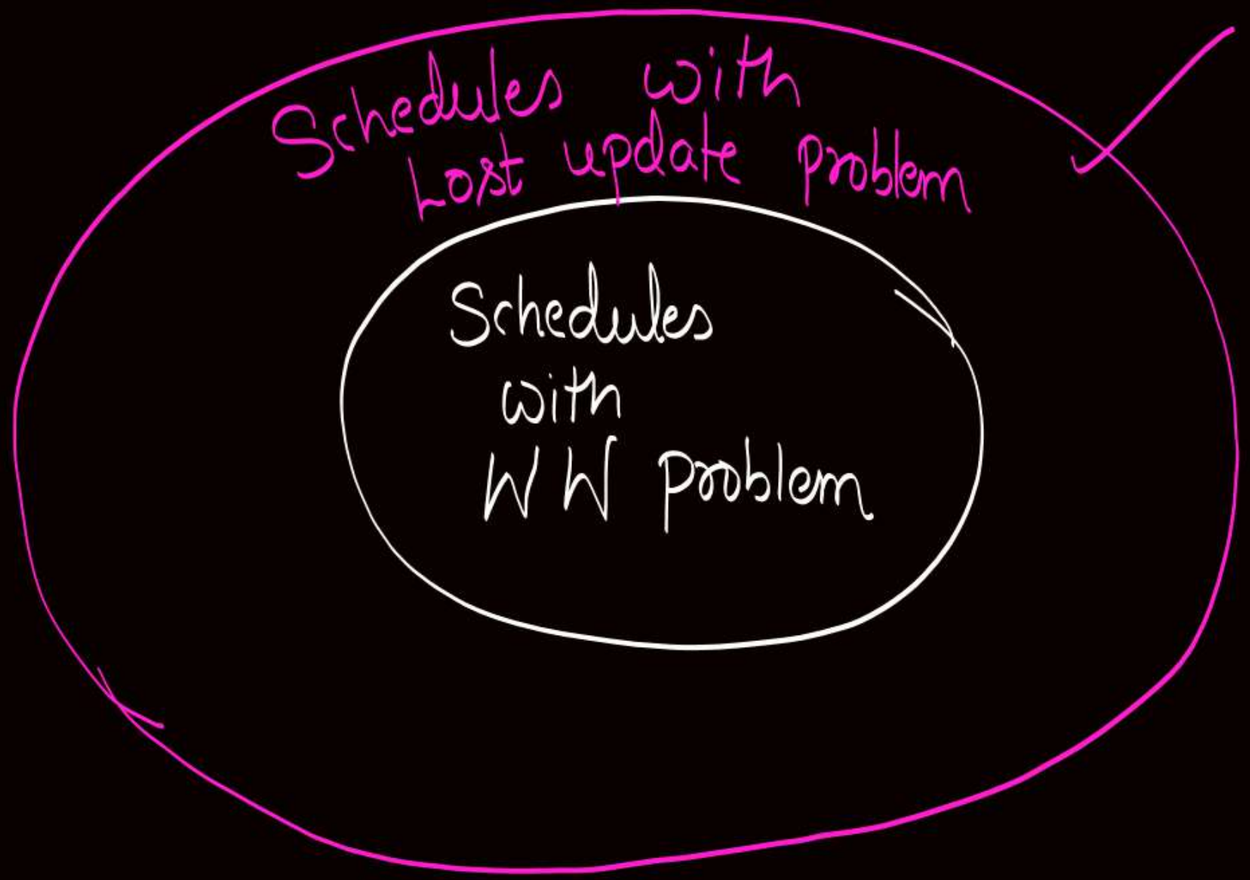
We can see that schedule S is a serializable schedule and equivalent serial schedule is $T_1 \rightarrow T_2$

let, ininitially the value of dataitem 'A' in the database = 100



∴ None of RW, WR and WW Problem are possible ⇒ but lost update problem is possible

Note:- Similar problem can exist in a non-serializable schedule as well for eg: Change $R_1(B)$ by $R_1(A)$


Schedules with
Lost update problem
Schedules
with
W W problem



2 mins Summary

- **Topic** Problems because of concurrent execution
- **Topic** RW problem
- **Topic** WR problem
- **Topic** WW problem
- **Topic** Lost update problem

THANK - YOU